

Chamarthi Om Naga Sai Mani Pavan

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EDUCATION

- **University of Illinois at Chicago** May 2025
Master's in Computer Science; CGPA: 3.5/4.0 Chicago, IL
Courses: Data Science, Database Systems, Big Data Mining, Query Processing in Database Systems, Computer Systems Security, Data Visualization & Analytics, Data Mining & Text Mining, Machine Learning, Advanced Linux Kernel Programming
- **Sir M Visvesvaraya Institute of Technology** May 2023
Bachelor of Engineering in Computer Science; CGPA: 8.4/10 Bangalore, India

TECHNICAL SKILLS

- **Languages:** Java, C, C++, Python, R
- **Web Technologies:** HTML, CSS, JavaScript
- **Database:** MySQL, MongoDB
- **Technologies:** Machine Learning, Mobile application development
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Tableau

EXPERIENCE

- **Java Developer Intern, Externs Club** Jan 2023 – Mar 2023
 - **Employee Management System:** Developed and maintained Java applications to manage employee records, role assignments, and payroll using Java Collections, Multithreading, and JDBC.
 - **Core Functionalities:** Implemented user authentication, data validation, and record management based on OOP principles.
 - **Agile Development:** Wrote JUnit test cases and debugged issues in an Agile environment for performance and maintainability.
- **Data Science Intern, Take It Smart** Aug 2022 – Dec 2022
 - **Machine Learning Models:** Developed and trained machine learning models using supervised learning techniques like Random Forest and XGBoost to predict outcomes such as house prices or customer churn.
 - **Feature Engineering:** Engineered features from raw data, applied scaling and encoding, and tuned hyperparameters using GridSearchCV to improve model accuracy.
 - **Model Evaluation:** Evaluated models using RMSE, R-squared, and accuracy to ensure generalization and prevent overfitting.

ACADEMIC PROJECTS

- **Air Bnb's Impact On Real Estate Trends:** Integrated Airbnb & Zillow data (280K+ records) using data cleaning, transformation, and zip code mapping to analyze Airbnb's impact on long-term rental trends. Built interactive D3.js visualizations (heatmaps, choropleths, bubble plots) using React & Leaflet, uncovering neighborhood-level insights on pricing and Airbnb density. Developed a predictive investment tool leveraging time-series analysis to generate "Invest/Don't Invest" recommendations based on Airbnb and rental price trends.
- **Price Prediction Using Zillow Data:** Developed a predictive model for house price estimation using Zillow data, leveraging advanced regression techniques like Gradient Boosting and Random Forest. Engineered key features and optimized model performance through hyperparameter tuning and cross-validation. Conducted EDA and visualized market trends using Python libraries to extract actionable insights.
- **Hospital Management System:** Engineered a Hospital Management System using PHP, MySQL, and HTML to manage healthcare data across Admin, Nurse, and Patient roles. Designed a normalized database schema for personnel, patients, vaccines, and scheduling. Built ingestion pipelines with real-time validations and transactional safety, supporting future analytics and reporting.

PUBLICATIONS

- **Conference Paper:** "Safe place suggestion using machine learning" – ITC affiliated conference (with Pabbati Jahnavi, Tadikonda Harshitha)
- **Journal Article:** "Safe place suggestion using machine learning" – Journal of Emerging Technologies and Innovative Research (JETIR), Vol. 10, Issue 5, May 2023

EXTRACURRICULAR ACTIVITIES

- **Head, University Events Management Team:** Led coordination and logistics for major university events, managing teams and resources.
- **Core Member, NGO – Abhi Kalpana:** Contributed to planning and executing community outreach programs and educational initiatives.